

Built from scratch using NumPy only

Problem Statement: Design and implement an ANN in Python to predict whether a customer will default on a credit card payment.

Dataset: Use a credit card dataset (Download from Kaggle)

Features: Credit limit, Age, Bill amount, Payment history, Previous payments, etc.

Target Variable: Default (0 or 1), 1 → Default; 0 → No default

Task 1: Data Preprocessing

Handle missing values/Encode categorical variables/Normalize/standardize features

Task 2: ANN Implementation

Implement: Weight initialization, Forward propagation, Backpropagation, Gradient descent

Task 3: Hyperparameter Tuning

Experiment with:

Learning rate: 0.001, 0.01, 0.1

Hidden layers: [n, 8, 1], [n, 16, 8, 1]

Optimizers: SGD & Adam

Compare performance of each combination

Task 4: Model Evaluation

Evaluate using: Accuracy, Confusion Matrix, Precision, Recall, F1-score

Task 5: Visualization: Plot loss vs epochs & Compare optimizers graphically

****Constraints****

No TensorFlow / PyTorch for core implementation, Only NumPy for ANN logic

Sklearn allowed only for evaluation

Expected Output:

Accuracy $\geq$ 80% (after tuning) & Comparative analysis of - Learning rates, Optimizers, Architectures

Extension Tasks: Implement ReLU activation, Add Dropout and Perform early stopping

At the end compare with: Logistic Regression & Decision Tree